higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## WELDING
## NQF LEVEL 2

(6031012)

**22 November 2019 (Y-Paper)**
**13:00–16:00**

**This question paper consists of 7 pages.**

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**TIME: 3 HOURS**
**MARKS: 100**

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Write neatly and legibly.

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## QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 The electrical current that is delivered by a welding machine during welding:

A Welding current
B Safety voltage
C Start current
D Load voltage

1.1.2 The component that prevents flames from entering the oxy-acetylene equipment:

A Gas hose
B Gas regulator
C Flashback arrestor
D Nozzle

1.1.3 Gas cylinders must be stored in the correct manner according to the safety requirements:

A In a wet area
B In a confined space
C In a laying position
D In an upright position

1.1.4 The acronym WPS stands for:

A Winding Process Specification
B Welding Procedure Specification
C Working Preventative Specification
D Welding Process Specialisation

1.1.5 Mechanical cutting equipment is called a:

A Metal cutting band saw machine.
B Plasma cutting machine.
C Oxy-fuel gas cutting machine.
D Bending brake machine.

(5 × 1) (5)

1.2 Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 The distance from the root to the face is known as the fillet weld throat.

1.2.2 The working pressure gauge on the regulators is used to check the amount of gas in the cylinder.

1.2.3 Safety harnesses are used when working at heights greater than three metres.

1.2.4 A heating box is used to keep electrodes dry.

1.2.5 Vertical welding is when the weld has to be performed horizontally.

(5 × 1) (5)
**[10]**

**QUESTION 2**

2.1 Protective equipment is very important during the welding process.

2.1.1 List FIVE types of protective equipment used during shielded metal arc welding. (5)

2.1.2 State ONE purpose of each type of protective equipment listed in QUESTION 2.1.1. (5)

2.2 Explain the process to follow in preparation for shielded metal arc welding. (6)

2.3 Give TWO corrective actions that must be taken on the two welding defects listed below:

2.3.1 Spatter

2.3.2 Porosity

(2 × 2) (4)

2.4 List FIVE safety measures for storing chemicals that are not in use. (5)
**[25]**

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## QUESTION 3

3.1 State step by step the assembly process of oxy-acetylene gas welding equipment. (5)

3.2 Draw a neat, labelled sketch of an unequal weld on a fillet joint in oxy-acetylene gas welding. (2)

3.3 FIGURE 1 shows a destructive test done on a butt joint in oxy-acetylene gas welding. Label the indicated parts. Write your answers next to the letters (A–C) in the ANSWER BOOK.

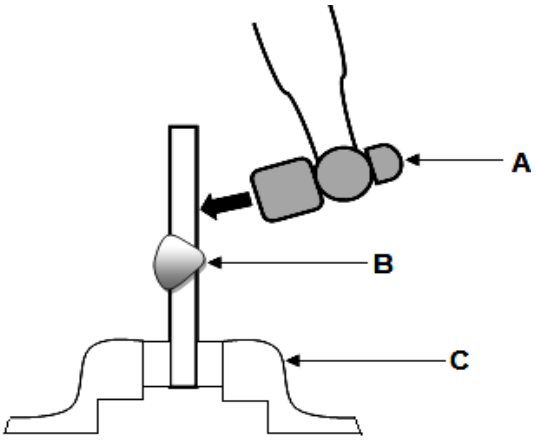



**FIGURE 1**

(3)
**[10]**

## QUESTION 4

4.1 Explain the cutting characteristics of mild carbon steel when using an oxy-acetylene cutting torch. (7)

4.2 State ONE purpose of the following components of oxy-acetylene cutting equipment.

4.2.1 Gas cylinders (1)

4.2.2 Content pressure gauge (2)

4.2.3 Gas hoses (1)

4.2.4 Flashback arrestors (2)

4.2.5 Working pressure gauge (2)

4.3 Name TWO types of defects that occur after the oxy-acetylene cutting process. (2)

4.4 Define the following terms for phenomena that occur during the oxy-acetylene cutting process:

4.4.1 Backfire

4.4.2 Flashback

(2 × 2) (4)

4.5 State TWO causes for the terms listed in QUESTION 4.4.1 and QUESTION 4.4.2. (4)

**[25]**

**QUESTION 5**

5.1 The TABLE below shows different welding codes and terms for different types of welding positions. List the welding codes and terms (5.1.1–5.1.5) and write your answers in the ANSWER BOOK.

| | |
|---|---|
| 1F | 5.1.1 … |
| 5.1.2 … | Horizontal welding position |
| 3F | 5.1.3 … |
| 5.1.4 … | Overhead welding position |
| Pipe welding at 45° position | 5.1.5 … |

(5 × 1) (5)

5.2 State the consequences of the incorrect assembly of gas metal arc welding equipment. (5)

5.3 State TWO types of gases used in gas metal arc welding. (2)

5.4 List the FOUR main aspects of preparing a work piece before welding can commence. (4)

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5.5 FIGURE 2 shows different types of components used in gas metal arc welding. State a function of the components listed as (5.5.1–5.5.4) under FIGURE 2 and write your answers in the ANSWER BOOK.

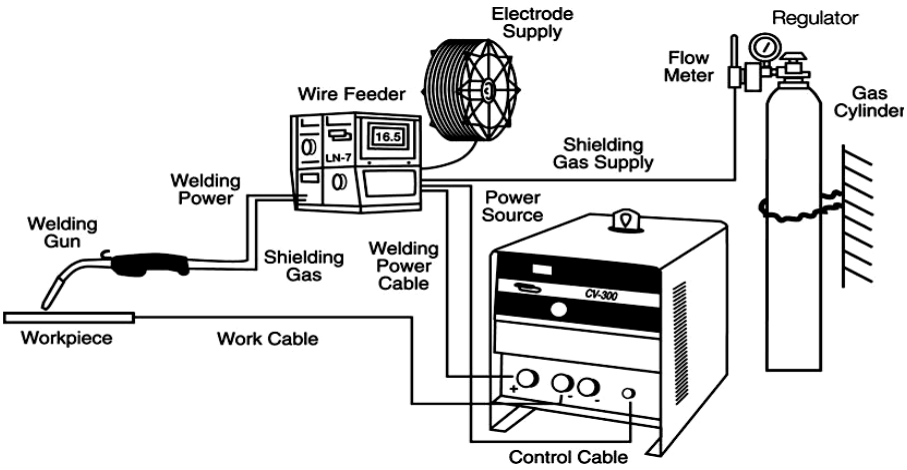



**FIGURE 2**

5.5.1 Wire feeder

5.5.2 Power source

5.5.3 Welding gun

5.5.4 Wire spool

(4 × 1) (4)
**[20]**

## QUESTION 6

6.1 Explain the process of adjusting a flame to a neutral flame in gas brazing. (5)

6.2 List FIVE safety measures when lashing gas cylinders. (5)
**[10]**

**TOTAL: 100**

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